

WEALTH TAXATION & LABOR SUPPLY: EVIDENCE FROM THE DANISH WEALTH TAX EXPERIENCE

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Abstract

I estimate the earnings response to the abolition of wealth taxation in Denmark, focusing on an efficiency argument to tax wealth advanced in the New Dynamic Public Finance literature. Under this argument, left to their own devices households would over-save and under provide labor upon the resolution of uncertainty. Answering the question is hard because

The argument to tax wealth holds that, under private information and stochastic earnings ability, an individual's optimal savings decision reduces their future labor supply incentives due to the absence of an insurance market. I Using a differences-in-differences strategy that doubled the threshold for wealth tax liability for couples in Denmark in 1989, I find a null labor income response, . . .

Using GMM estimator, I estimate consumption Euler equations to rule between a lifecycle model with an endogenously missing market due to moral hazard and a full insurance benchmark. I find that while the theoretical mechanism is plausible, its empirical relevance for Denmark, and potentially other developed economies, is limited. Using a differences-in-differences strategy, I find a null

JEL: J22, H21, H31, E21, E24.

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1 INTRODUCTION

Recent years have seen a surge in political and academic interest around wealth taxation, spurred by growing economic inequality and calls for policy tools to address wealth concentration. While much of this debate has focused on redistributive goals, a gap remains in our understanding of the broader economic impacts of wealth taxation – particularly, its impact on labor earnings (Scheuer & Slemrod 2021). This paper addresses a key question in public finance: how do wealth taxes influence household labor supply decisions, including those of the primary and secondary earners within a household, as well as decisions related to retirement and self-employment? Understanding this relationship is crucial for designing tax policies that balance equity and efficiency, especially because multiple theories of economic behavior defend a Pareto-improving wealth tax to incentivize labor supply (Goloso et al. 2006).

To address this literature gap, this paper leverages multiple quasi-experimental designs arising from Denmark’s wealth tax experience and utilizes comprehensive administrative registers to empirically analyze the effects of wealth taxation on labor supply decisions, retirement timing, and self-employment activities. I find that the abolition of the Danish wealth tax led to a small labor earnings response for households where both earners are employed laborers. The gross household income (including income from business activities) elasticity for households with any self-employed member is statistically significant and equal to VALUE. Along the margin of business income from self-employment including net profits, the estimated elasticity is even larger. Potentially talk about the no of employees of self-employed people with workers.

The Danish context offers several advantages: the wealth tax was imposed uniformly across the country, minimizing concerns about geographic mobility influencing the results;

detailed third-party reporting reduces concerns related to tax evasion and misreporting; and the richness of the data allows for an examination of various labor supply margins. By exploiting these features, I provide robust causal evidence on how wealth taxes influence a range of labor market outcomes, including for the self-employed. An additional benefit in studying the Danish experience stems from Denmark being a small open economy. The specific argument that a wealth tax corrects incentives to provide labor income is less potent in a closed general equilibrium economies with real interest rate adjustments. This is because a tax on wealth will be followed by a countervailing real interest rate response, limiting the Pareto-improvements of a wealth tax ([Farhi & Werning 2012](#)). Because Denmark is a small open economy, the countervailing force of real interest rate adjustment is unlikely to materialize in response to a wealth tax cut or abolition.

This paper contributes to the literature on wealth taxation and labor economics along several dimensions. First, I document the behavioral labor earnings response following the abolition of the Danish wealth tax using a differences-in-differences identification strategy. The Danish wealth tax had one of the most expansive wealth tax bases globally in terms of fraction of households liable. Hence, we are best placed to capture labor supply incentives for treated households. I contrast the behavior of households subject to the wealth tax (treatment group) to households who are not but are nonetheless within the top 5% of the wealth distribution (control group). I validate my empirical findings by employing a documented identification strategy that was previously used by [Jakobsen et al. \(2020\)](#) to identify the effect of wealth taxes on wealth accumulation, comparing coupled households (treatment group) to singles (control group) following a doubling of the tax exemption threshold for couples.

Second, by linking the number of employees in the main business of self-employed wealth taxpayers, this paper is the first to document that wealthy self-employed households respond

to a wealth tax abolition by reducing their number of employees due to their reduced incentives to work. This provides an estimate of the trickle-down effects of improving the incentives to work at the top of the wealth distribution. While [Risch \(2023\)](#) documents a similar channel using corporate income tax rate changes in the US to study its pass-through effect on firm employees, he finds no change in employment following a corporate income tax increase.

Third, by documenting the reduction of employment in firms whose self-employed owners are liable to the wealth tax, this paper speaks to the optimal tax literature documenting an additional relevant channel of adjustment to wealth taxes that might dominate previous rationales to tax wealth. Arguments to tax wealth usually relate to distributional concerns arising from an unequal distribution of wealth or to the implications of taxing wealth on own-labor supply outcomes ([Piketty & Saez 2012](#), [Shourideh 2013](#), [Rogerson 1985](#), [Farhi & Werning 2012](#)). If we focus on the efficiency arguments to tax wealth, this paper provides an additional reason to tax wealth: the labor effort expended to generate income at the top of the distribution is complementary to labor effort across the rest of the labor income distribution. This is particularly relevant in the context of New Dynamic Public Finance models, which argue that wealth taxation can counteract inefficiencies arising from private information and stochastic earning abilities. In those private information models, left to their own devices households would over-save in order to hedge against the uncertainty of their future earnings profile. This then reduces their labor supply incentives on the resolution of uncertainty. Wealth taxation thus acts to dis-incentivize Pareto-dominated over-saving and correct labor supply decisions. To the best of my knowledge, there is no existing work – neither theoretical nor empirical – on the trickle-down benefits to workers at the bottom of the labor income distribution of a wealth tax on wealthy self-employed owners by Pareto-improving their incentives to work..

Fourth, by studying the abolition of the wealth tax and contrasting household labor supply responses with an earlier reform to the liable tax base, I contribute to the literature on how perceptions of tax reform permanence affect household behavior. I find that the effect of the wealth tax abolition on labor supply and gross income generation is larger and more persistent. In the case of Denmark, abolishing the wealth tax meant that there was a significant administrative hurdle to reintroduce it because Denmark stopped collecting the necessary information to administer the tax. In contrast, during the earlier quasi-experiment in 1989, the exemption threshold for coupled households was doubled to achieve horizontal per capita equity with single-individual households.

More generally, the paper falls into a nascent literature studying the effects of wealth taxation on household behavior ([Seim 2017](#), [Brülhart et al. 2019](#), [Jakobsen et al. 2020](#), [Ring 2024](#)). To that literature, I emphasize the differential labor responses between main and secondary earners within a household. I find **VALUE**.

Two puzzles remain in documenting the impact of the wealth tax on the joint distribution of earnings and savings decisions. I leave those for future work. Firstly, given that household gross income fell in response to the reform, how do we rationalize this finding with the finding of rising wealth levels in response to the tax cut? This work would compare life-cycle models of behavior under private information economies compared to economies with complete information. Secondly, more theoretically, what are the optimal wealth tax formulas given the empirical relationship documented here between self-employed wealthy business owners and laborers in their firms?

2 THE DANISH WEALTH TAX EXPERIENCE

Denmark imposed an annual wealth tax on taxable net worth until and including 1996 on households at the top of the wealth distribution. Most assets (including bonds, cash, deposits, equities, housing, and business assets) held by households were subject to the tax with the notable exception of pension wealth. [Jakobsen et al. \(2020\)](#) document much of the details related to the wealth tax. However, I reiterate some of the most important aspects of the tax program for completeness.

One marginal tax rate applied to all taxable wealth above an exemption threshold. Figure 3 is a scatterplot of the evolution of the marginal tax rate over time and the relevant exemption thresholds that applied. Over the period, there was a trend reduction in the marginal rate that applied. This occurred over two periods: firstly over 1989-1992, the rate was halved from 2.2% to 1%; and secondly, over 1995-1997, the tax was completely abolished.

During the same period, there was trend increase in the nominal exemption allowance that applied from around 1.25M Danish Krone (DKK) to around 2.4M DKK. Additionally, over 1989-1992, the exemption threshold that applied for married households was doubled to achieve horizontal equity with single-person households. This suddenly excused a fraction of moderately wealthy taxpaying households.

Much of the net worth items that make up taxable wealth are reported by third parties such as financial institutions. This includes for example bonds, equities, and housing assets. However, some items are self-reported including cash, unlisted equities, assets held abroad, and non-corporate business assets, allowing for potential evasion responses through under-reporting these assets on tax returns and then making gains on selling their appreciated on the abolition of the wealth tax.

I employ three Identification strategies to study the effect of wealth tax changes on labor supply incentives using reforms that occurred in the 1980s and 1990s in Denmark. I detail the wealth tax changes and how they can be potentially used to distinguish between the different informational regimes.

- Change 1: relies on a 1989 reform that doubled the threshold for which couples start being liable for paying a wealth tax. This was done to be more fair for couples. Therefore couples within a threshold suddenly had their 2.2% wealth tax reduced to zero.
- Change 2: the tax on wealth was also reduced over 3 years between 1989-1991 from 2.2% to 1%.
- Change 3: the wealth tax was abolished in 1997.

Potential ID strategies:

- ID1: Using the first policy change, define the treated households with in the old wealth threshold but outside the new wealth threshold. Define the control group as singles in the same wealth threshold who continued to pay the tax. Treatment period is 1989.
- ID2: Using the first policy change, define the treated households with in the old wealth threshold but outside the new wealth threshold. Define the control group as couple just below the original wealth threshold. Treatment period is 1989.
- ID3: Denmark operates a horizontal upper tax cap. Use households above the horizontal upper cap as part of a control group around 1989. This is because their marginal wealth tax is zero. Individuals just below the threshold pay a wealth tax of 2.2% before 1989. This was then reduced to 1% through to 1991.

- ID4: Apply ID3 to the abolition of the wealth tax in 1997 in Denmark.

ID1-3 were adopted previously by [Jakobsen et al. \(2020\)](#) to study the effect of wealth tax on wealth accumulation. ID4 was not used previously since the abolition of the wealth tax was followed by a break in the measurement of wealth in the administrative data. Since I do not look at wealth outcomes but rather labor market choices at the top of the distribution of wealth, I am able to identify effect of the wealth tax abolition on labor supply decisions.

3 THE IMPORTANCE OF WEALTH TAXPAYERS' LABOR DECISIONS

Why is the Danish data of interest?

The data is useful for multiple purposes. Individual income and wealth are observed over a long period of time. Population-level coverage implies gains from observing responses even at the top of the distribution of wealth. Another key positive in favor of the data is the presence of convincing instruments due to the wealth tax reform in 1988 ([Jakobsen et al. 2020](#)). Evidence from the joint distribution of income, wealth, and consumption can shed light on competing theories in public finance.

What is exactly needed from the data?

[Karaivanov & Townsend \(2014\)](#) conduct their analysis in Thai villages on entrepreneurial households, who consume, work, and invest (in their firm). [Karaivanov & Townsend \(2014\)](#) run their estimates under different assumptions regarding the data availability and test the alternative model specifications. In the first instance, they estimate the competing models using the joint distribution of business assets, investment, and income data. They then replicate their analysis with consumption and income data alone. Then finally, they use data on business assets, consumption, investment, and income all-together. Based on maximum likelihood and other goodness of fit criterion, they conclude which model(s) fit the data best. A similar analysis in the Danish administrative data would require data at least two of the following: income data, consumption data, and wealth data.

The additional gain from using the Danish data would be accessing the labor income and assessing how taxable income responds to changes in the wealth tax. This can then be related to potential responses from the dynamic Mirrleesian model and that of the full insurance benchmark.

The size of the wealth tax corresponds closely to estimates of the theoretically optimal tax on wealth. It is also the case that this is one of the largest wealth taxes being removed.

Concerns

- *Old Individuals and their labor response?*

One concern raised is that the labor response might be zero due to the presence of many retired individuals in the treated subsamples. This is one way to think of it. An alternative way is to think of the retired individuals as individuals with a certain $\theta_t = 0 \quad \forall t > t_r$ where t_r is the age of retirement. For those individuals, we are certain the inverse Euler equation doesn't apply since their ability measure is public information. Hence, assuming retirement is an absorbing state (as assumed in some papers in public finance ([Goloso & Tsyvinski 2006](#))), older individuals might be a good reference sample compared to individuals who work and hence have the option of this behavioral adjustment channel. This of course would have to assume that mortality risk is not a concern and that retirement decisions are exogenously fixed.

- *Rich individuals and their labor response?*

I share this concern that the meaningful margin of response might not be the labor

supply margin for the wealth tax reforms conducted in Denmark. One way to attempt at dealing with this would be to estimate the labor responses at different wealth levels for those affected by the reform vs. not.

Another way to address this concern would be to use self-employment and own-business earnings as a way to approach the theoretical construct of individual ability to generate labor income. Ultimately, the theory is concerned with the effort exerted by individuals to generate income. This effort is what faces dynamic distortions if policy-makers do not tax wealth.

4 INTENTION TO TREAT EFFECT OF WEALTH TAX REFORMS ON HOUSEHOLD EARNINGS

4.1 ABOLITION DIFFERENCES-IN-DIFFERENCES

The abolition occurred in 1997. Households who were once liable to pay the wealth tax no longer had to. Based on the sample of households who were wealth taxpayers for the 3 years prior to the reform, that is 1994-1996, I construct a group that was treated by the reform. I compare this group to households whose wealth did not exceed the exemption threshold but were nonetheless in the top 5% of the wealth distribution in Denmark during each of the years 1994-1996. this group to Regression of Interest:

$$L_{it} = \sum_{j \neq 1996} \delta_j \cdot Year_j \cdot Treat_{it} + \gamma_i + \eta_t + X_{it} + v_{it} \quad (1)$$

L_{it} is a measure of labor supply. A list of examples are:

- Logged household taxable labor income.
- Individual and Spousal retirement decisions.
- Measure of self-employment and income from self-employment (conditional on being self-employed).

Assuming parallel trends hold and if $\delta_{ITT} = 0$, the economic significance of the labor supply mechanism to tax wealth is weak.

5 DATA EXPLORATION BASED ON REDUCED FORM

SECTION

(i) Specification 1

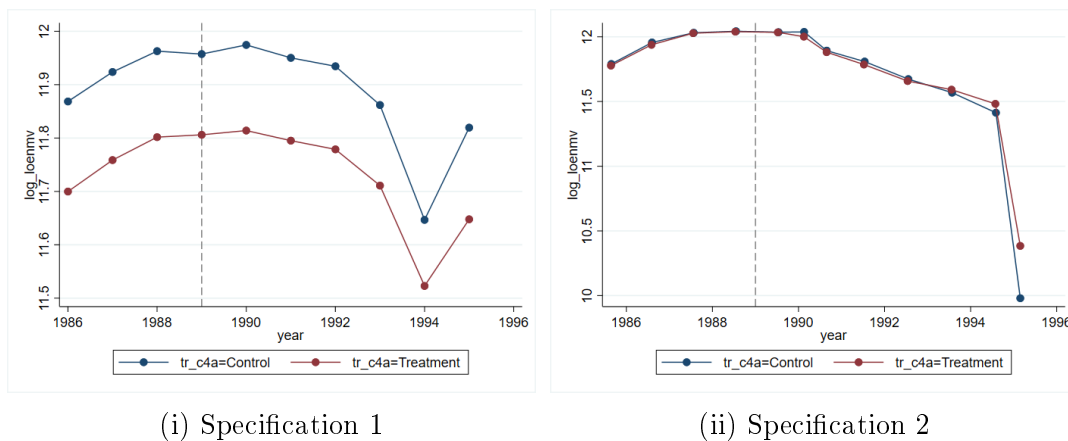
(ii) Specification 2

Figure 1: Logged household labor income using ID1.

Notes. Treated households refers to married households whose marginal wealth tax rate fell from 2.2% to 0% in 1989. Control households refers to singles whose wealth tax rate continued to be 2.2% in 1989 and fell to 1% through to 1991. Specification 1 refers to a Diff-in-Diff specification without household-level fixed effects. Specification 2 adds household-level fixed effects.

This early exercise indicates that the wealth tax changes have little effect on labor incomes. Pending confirming this along multiple margins of response and using the other identification strategies, this might lead us to believe that the New Dynamic Public Finance argument to tax wealth does not really matter in practice.

6 RESULTS



(i) Specification 1

(ii) Specification 2

Figure 2: Logged household labor income using ID2.

Notes. Treated households refers to married households whose marginal wealth tax rate fell from 2.2% to 0% in 1989. Control households refers to married households who were below the wealth tax threshold and paid 0% wealth tax in 1989 but were part of the top 5% of wealth in 1989. Specification 1 refers to a Diff-in-Diff specification without household-level fixed effects. Specification 2 adds household-level fixed effects.

Table 1: Summary Statistics on Main Analysis Sample

Elasticities: ϵ wrt $(1-\tau_w)$ Dependent Variables	1989 Wealth Tax Reform				1997 Wealth Tax Abolishment	
	Couples within vs. Couples Below		Couples Within vs. Singles Below		Tax-payers vs non-payers	
	(1)	(2)	(3)	(4)	(5)	(6)
HH Labor Income	-.00217 (.00202)	-.00241 (.0021)	-.00301 (.00232)	-.00208 (.00523)	-.00233 (.00534)	-.00153 (.00482)
2 nd Earner's Labor Income	-.00217 (.00202)	-.00241 (.0021)	-.00301 (.00232)	-.00208 (.00523)	-.00233 (.00534)	-.00153 (.00482)
HH Total Income	-.00217 (.00202)	-.00241 (.0021)	-.00301 (.00232)	-.00208 (.00523)	-.00233 (.00534)	-.00153 (.00482)
2 nd Earner's Total Income	-.00217 (.00202)	-.00241 (.0021)	-.00301 (.00232)	-.00208 (.00523)	-.00233 (.00534)	-.00153 (.00482)
Retired (Main Earner)	-.00217 (.00202)	-.00241 (.0021)	-.00301 (.00232)	-.00208 (.00523)	-.00233 (.00534)	-.00153 (.00482)
Retired (2 nd Earner)	-.00217 (.00202)	-.00241 (.0021)	-.00301 (.00232)	-.00208 (.00523)	-.00233 (.00534)	-.00153 (.00482)
Self employed (Main Earner)	-.00217 (.00202)	-.00241 (.0021)	-.00301 (.00232)	-.00208 (.00523)	-.00233 (.00534)	-.00153 (.00482)
Self employed (2 nd Earner)	-.00217 (.00202)	-.00241 (.0021)	-.00301 (.00232)	-.00208 (.00523)	-.00233 (.00534)	-.00153 (.00482)
HH Total Income	-.00217 (.00202)	-.00241 (.0021)	-.00301 (.00232)	-.00208 (.00523)	-.00233 (.00534)	-.00153 (.00482)
2 nd Earner's Total Income	-.00217 (.00202)	-.00241 (.0021)	-.00301 (.00232)	-.00208 (.00523)	-.00233 (.00534)	-.00153 (.00482)
<i>Matching</i>	N	N	N	N	N	N
<i>Controls</i>	N	N	N	N	N	N
<i>HH Fixed Effects</i>	N	Y	N	Y	N	Y

Notes.

7 CONCLUSION

Debates on Wealth taxation gained renewed interest. Recently, two American presidential candidates backed high wealth tax rates on the very richest Americans. In fact, opinion polls suggest that Americans would vote for a tax on the wealth of the superrich.

Leading academic arguments to tax wealth from the personal tax code usually rest on strong normative assumptions, such as taxes on inheritance ([Farhi & Werning 2013](#)) and redistribution of capital income risk ([Shourideh 2013](#)).

This paper sets aside normative rationales for wealth taxation and investigates empirical evidence for a key positive argument to tax wealth in the literature. It aims to answer the question of whether the presence of private information and stochastic evolution of individual ability is corroborated in a developed economy context. If so, it aims to answer the question on the size of the earnings response caused by a change in the wealth tax. The paper would contribute to the policy debate surrounding wealth taxes. It would thus influence the academic literature by assessing the need for better positive theories of capital taxation.

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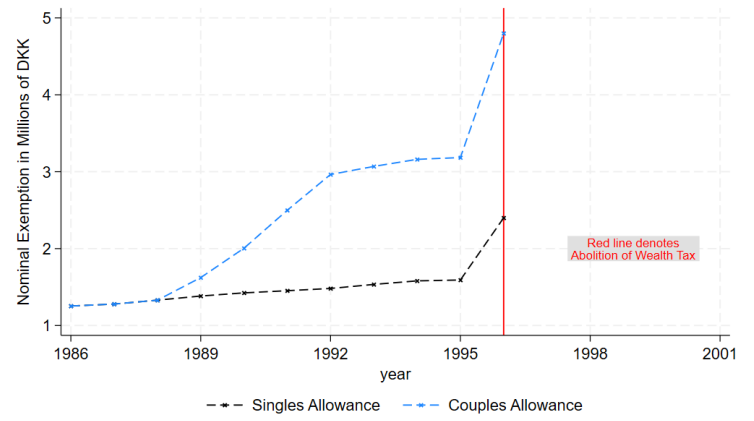
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8 FIGURES AND TABLES

Figure 3: Evolution of Wealth Tax Marginal Rate and Exemption Threshold



(i) Marginal Tax Rate



(ii) Allowance by Marriage Status

Notes. This figure is calculated based on work by [Jakobsen et al. \(2020\)](#). The red line in [3ii](#) is marked at the year 1996 because that is the last year where there was an exemption allowance. Press [here](#) to go back to the main text.

Table 2: Summary Statistics for the Main Analysis Sample

	Couples DD: 1989				Abolition DD: 1997	
	Full Population	Couples within (Treatment)	Singles within (Control)	Couples Below (Control)	Households Above (Treatment)	Households Below (Control)
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Characteristics</i>						
Taxable Wealth	749,373 (3768111)	3,202,171 (570633)	3,196,087 (575043)	2,071,910 (176544)	8,876,322 (34488100)	2,033,292 (284487)
Asset Share in Housing	0.44 (54.04)	0.35 (0.27)	0.31 (0.26)	0.51 (0.31)	0.26 (0.18)	0.50 (0.25)
Asset Share in Equity	0.02 (2.61)	0.07 (0.13)	0.08 (0.15)	0.03 (0.07)	0.07 (0.14)	0.03 (0.07)
Fraction Self-Employed: Main Earner	0.05 (0.22)	0.06 (0.24)	0.28 (0.45)	0.06 (0.23)	0.13 (0.34)	0.10 (0.30)
Fraction Self-Employed: Second Earner	0.10 (0.30)	0.41 (0.49)	0.28 (0.45)	0.27 (0.45)	0.33 (0.47)	0.20 (0.40)
Years of Education	12.16 (3.17)	12.37 (3.34)	12.59 (3.72)	12.16 (3.22)	13.36 (3.07)	12.61 (3.28)
Age	49.76 (16.04)	58.56 (10.51)	62.62 (12.30)	57.82 (10.41)	58.50 (12.48)	61.10 (11.89)
<i>Outcomes</i>						
HH Labor Income	260,647 (281740)	255,225 (375009)	93,759 (189782)	281,928 (338498)	338,702 (681595)	215,112 (332197)
Labor Income: 2nd Earner	123,681 (147761)	57,515 (119712)	93,759 (189782)	68,728 (119337)	105,105 (222130)	77,530 (152325)
Gross HH Income	405,235 (371962)	708,307 (403383)	441,100 (251623)	558,073 (292620)	1,044,916 (2247105)	520,060 (308557)
Gross Income: 2nd Earner	203,088 (170405)	182,074 (128892)	441,100 (251623)	159,837 (114466)	389,499 (593299)	244,389 (166810)
Fraction Retired: Main Earner	0.27 (0.45)	0.24 (0.42)	0.38 (0.49)	0.25 (0.43)	0.30 (0.46)	0.43 (0.50)
Fraction Retired: Second Earner	0.25 (0.43)	0.21 (0.40)	0.38 (0.49)	0.21 (0.41)	0.29 (0.45)	0.42 (0.49)
Number of Households	1,308,465	21,963	7,410	15,790	33,908	39,966

Notes. The table reports means and variances for the different treatment arms along with the general population of households in the sample for 3 years prior to the reform. For the Couples DD, this corresponds to the years 1986-1988. The Couples DD treatment occurs in 1989. For the Abolition DD, this corresponds to the years 1994-1996. The Abolition DD occurs in 1997. Standard Deviations are reported in brackets under each row. The sample consists of a balanced panel of households for 11 years in the case of the Couples DD and for 8 years in the case of the Abolition DD, excluding households where both individuals are younger than 20 in any of the years. Couples Within are households whose exemption threshold was doubled in 1989 and whose wealth was within the old threshold and the new threshold for the three years prior to the reform (and were thus not liable for the wealth tax). Singles Within are single-person households within the same two thresholds for the three years prior to the reform (and hence don't benefit from a reduced wealth tax bill). Couples Below are an additional control group corresponding to coupled households below the old exemption threshold for 3 consecutive years and are in the top 5% of households in the wealth distribution. Households Above are households who were above the exemption threshold for 3 consecutive years prior to the abolition of the wealth tax. Households Below are households who were below the exemption threshold for 3 consecutive years prior to the abolition of the wealth tax and are in the top 5% of households in the wealth distribution. The Full Population sample is inclusive of the different treatment arms and provides summary statistics on the 1986-88 period. Taxable wealth refers to net worth as defined by the Danish Tax Authority, this measure of wealth notably excludes retirement wealth. HH labor income refers to total household income from supplying labor including fringe benefits, tax-free salary, anniversary and severance pay, and value of stock options. Gross HH Income refers to taxable income generated by a household including labor income and capital income removing some deductions. Capital income includes interest income, rental income, and business capital income minus interest on debt. Main earner is defined by the spouse in the household with more gross income. Appendix C provides more information on variable definitions. All monetary amounts are reported in 2017 DKK (exchange rate was 6.2 DKK/USD). Press [here](#) to go back to the main text.

A OVERVIEW OF EFFICIENCY GROUNDS TO TAXING WEALTH UNDER NEW DYNAMIC PUBLIC FINANCE

An important argument in the literature to tax wealth relies on an informational inefficiency that diminishes the incentive of individuals to supply labor at the optimal level. Taxing wealth hence corrects incentives to supply labor. In this “new dynamic public finance” argument, the presence of a moral hazard informational inefficiency, due to the stochastic evolution of an agent’s ability and its lack of observability by a planner of an individual’s ability to generate labor income, leads to an endogenously incomplete market. In this world, the optimal decision for an individual is to make consumption and labor supply plans according to the inverse Euler equation, as opposed to the standard Euler Equation (which solves the optimal allocation problem of the planner). The two Euler equations are listed as follows,

$$E\{\beta(1 + r_{t+1})U'(C_{t+1}(\theta_{t+1}))\} = U'(C_t(\theta_t)) \quad \text{Planner’s Euler Equation}$$

$$\frac{1}{\beta(1 + r_{t+1})} E_t \left[\frac{1}{U'(C_{t+1}(\theta_{t+1}))} \right] = \frac{1}{U'(C_t(\theta_t))} \quad \text{Inverse Euler Equation}$$

By Jensen’s inequality, the inverse Euler Equation implies the need for a intertemporal wedge (tax) on wealth. Intuitively, as opposed to a full information benchmark, left to their own devices, individuals would save too much relative to the constrained pareto optimal allocation. Alternatively, the desired labor supply at the optimal allocation is incompatible with free savings. This is because an increase in wealth has two effects on welfare in the following period. Although it increases the resources available for the individual in the second period, it has negative impact on labor supply incentives. This paper provides evidence on whether this argument to tax wealth is relevant for policy design.

Suppose we live in an economy with private information, stochastic ability, and incomplete markets, what should this wealth tax be to achieve the constrained pareto optima? Multiple papers in the macro-public finance tradition attempt to provide an answer for this question using quantitative exercises in a partial or general equilibrium framework. Under standard parametric assumptions, if consumption log-normally distributed with a variance σ_ϵ^2 , [Farhi & Werning \(2012\)](#) characterise the optimal capital distortion as

$$\tau_t^K = 1 - \exp(-\sigma_\epsilon^2) \quad \text{Optimal Wealth Distortions}$$

so that $\tau_t^K \approx \sigma_\epsilon^2$, when σ_ϵ is small. Empirical estimates, using the panel study of income dynamics, of the size of σ_ϵ^2 suggest values between 0.56% and 1% (Deaton & Paxson 1994, Blundell et al. 2008, Heathcote et al. 2014). Conesa et al. (2009) in another quantitative exercise, using a dynamic general equilibrium model, estimate the optimal capital income tax to be around 36%, which translates to 2% if the tax was implemented on capital as opposed to capital income. These estimates are close to the magnitude of the wealth tax change studied in this paper.

A.1 FURTHER EMPIRICAL AND THEORETICAL LITERATURE CONNECTIONS

Work on the effect of wealth taxes on labor supply outcomes is nascent and doesn't address the efficiency implications of taxing wealth. However, a recent literature that focuses on the impact of wealth taxes on individual behavior emerged, owing to increasingly available high quality data on the joint distributions of income, consumption, and wealth. Brülhart et al. (2019) study a reduction in the wealth tax in one Swiss canton relative to another. They find no labor earnings response to a reduction in wealth taxes. In their study, wealth is self-reported to the tax authority leaving a lot of room for evasion concerns. Ring (2024) studies a change in the tax assessment of housing wealth in Norway that differentially affected household's estimated wealth. He finds that an increase in exposure to wealth taxes led to increased labor earnings. One concern with this procedure of identifying the effect is that housing provides a consumption good. Hence, exposure to a tax on housing wealth might not alter the individual's intertemporal decisions. In both Brülhart et al. (2019) and Ring (2024), an additional concern with the estimates is the identification argument relies on location decisions and assumes their rigidity. Seim (2017) also investigates the impact of a wealth tax reform using Swedish Administrative data. His focus is on the savings behavior and evasion responses. Separately, Jakobsen et al. (2024) document migration responses to wealth taxes in Sweden and Denmark and find small responses.

The Danish wealth tax reform informs the literature on the impact of wealth taxation on labor supply in multiple ways. Firstly, the Danish wealth tax reforms were not tied to locations. Hence, location choices within Denmark would not be a concern for identification. Secondly, the third party reporting of most assets reduces the evasion responses of individuals. Thirdly, the availability of data allows me to study responses to the tax cut at different margins of labor supply like spousal labor supply, self-employment, and early retirement

decisions. Although one concern is that the wealth tax was paid only by the top 2-3% of the distribution, reducing scope for a labor supply response, it is likely that a reduction in wealth tax (like the Danish experiment) will reduce the incentives to work for those adults most, since their wealth can generate enough returns for them to not need labor income. Additionally, this paper would be the first to only consider the labor supply response to wealth taxation uniquely as opposed to papers that focus more on savings behavior and wealth accumulation. ([Jakobsen et al. 2020](#), [Seim 2017](#))

Despite a large literature in the macroeconomics and public finance theory traditions, past empirical literature comparing an endogenously incomplete market informational inefficiency to a self insurance benchmark is little ([Attanasio & Weber 2010](#)). In fact, empirical literature assessing this as an argument to tax wealth is non-existent as far as I am aware. This probably owes to two reasons. Firstly, data on wealth is extremely rare and often unreliable. Secondly, until recently, computational power put severe constraints on assessing these competing theories. Multiple papers however attempted to deal with the technical challenges and tested models of endogenously incomplete markets against those of complete markets. However, those were mainly conducted in developing country contexts. [Ligon \(1998\)](#), concerned with risk-sharing in three village economies in India studied by [Townsend \(1994\)](#), uses data on consumption to answer whether the standard Euler equation or the inverse Euler equation applies. In doing so, [Ligon \(1998\)](#) tests between three different models (Private Information, Permanent Income, and Full Insurance) using GMM, imposing restrictions on the consumption behavior implied by the alternative models. His findings favor a private information world for two of the villages and permanent income hypothesis for the final one. The study is naturally not concerned with labor supply decisions nor wealth taxation. It aims to quantify the extent to which social/kinship-based insurance is substituting for government in a developing country context. A further body of work in that tradition followed suit. [Kinnan \(2021\)](#) studies consumption and risk-sharing in similar context in Thailand. Surprisingly, in those studies, there is no interest in the interaction of these effect with labor supply incentives and decisions.

A wider empirical literature concerned with consumption and risk-sharing exists (See [Attanasio & Weber \(2010\)](#) for a summary). In the UK, [Attanasio & Pavoni \(2011\)](#) explain the excess smoothness of consumption ([Campbell & Deaton 1989](#)), using models of moral hazard similar to the dynamic Mirreleesian model. Using US data, [Blundell et al. \(2008\)](#) find evidence of some partial insurance against permanent income shocks. However, as

[Attanasio & Weber \(2010\)](#) conclude research on models with asymmetric information and their implications “is only beginning.” Empirical studies that attempt to use wealth tax changes to answer questions within the framework of these models are entirely non-existent. This is while the dynamic public finance agenda has achieved significant progress that yields testable predictions and have implications for capital taxation ([Golosov & Tsyvinski 2015](#)).

Methodologically, [Karaivanov & Townsend \(2014\)](#) provide and apply a method to compare across multiple informational regimes and financial structures. Based on maximum likelihood and linear programming, their approach is categorised by optimising the model subject to the original incentive compatibility constraints rather than using the common first-order approach (used by [Rogerson \(1985\)](#) and [Ligon \(1998\)](#) in public finance applications). This lead to an environment that allows for measurement error, unobserved heterogeneity, and non-convexities in the data. Another attractive feature of their method is its ex ante agnostic about the underlying data generation process. Although the application focuses mostly on financial constraints and their implications for insurance in developing Thailand. The method can be employed in the current exercise. The costs of this methodology is the curse of dimensionality that immediately bites as grids of state variables become more fine. The added benefit that this paper can bring-forth is that there are wealth tax changes that can imply ‘clean’ behavioral labor supply responses, coupled with an assessment of the inverse Euler equation argument to tax wealth.

B [LIGON \(1998\)](#) EXERCISE: DISTINGUISHING PERMANENT INCOME AND PRIVATE INSURANCE

This appendix details work-in-progress assessing directly which of the competing models of household behavior apply – a finding in support of the self-insurance or complete markets would be consistent with the reduced form evidence in the paper and find limited support for taxing wealth on efficiency grounds related to the supply of labor. The exercise follows [Ligon \(1998\)](#). Here, I distinguish whether a self-insurance model explains the data better or a private information model. The test would fail to provide implications under full insurance. Assuming $\beta * R = 1$ and $U(c) = c^{1-\gamma}/(1-\gamma)$, where c is consumption, the difference between a self-insurance model and a model with private information can be described by estimating

b_0 in the following equation, using GMM:

$$E_t \left[\left(\frac{c_{i,t+1}}{c_{i,t}} \right)^{b_0} - 1 \right] = 0 \quad (\text{Test of the Euler Equation})$$

Suppose we observe a sample c_{it} for $t = 1, \dots, T - 1$ years and for $i = 1, \dots, N$ households. Separately on a pre-reform treated sample, post-reform treated sample, pre-reform control sample, and a post-reform control sample, do the following:

- Estimate b_0 by the Generalised Method of Moments (GMM).

$$E \left[\left(\frac{c_{i,t+1}}{c_{i,t}} \right)^{b_0} - 1 \right] = 0 \quad (2)$$

- Next period's consumption is endogenous, instrument using $Z_{it} \in$ past income, past wealth.
- Estimate b_0 and J -statistic by GMM. Under rational expectations, the identifying restrictions are given by:

$$E \left[\left(\left(\frac{c_{i,t+1}}{c_{i,t}} \right)^{b_0} - 1 \right) \cdot Z_{it} \right] = 0 \quad (3)$$

where the equation of interest is:

$$E \left[\left(\frac{c_{i,t+1}}{c_{i,t}} \right)^{b_0} - 1 \right] = v_{t+1} \quad (4)$$

- If $b_0 < 0$ at the relevant significance level. Then infer that data was generated by permanent income hypothesis.
- If $b_0 > 0$ at the relevant significance level. Then infer that data was generated by private information economy.

C DATA DEFINITIONS

D ROBUSTNESS

D.1 ROBUSTNESS TO CHOICE OF PRE-REFORM YEARS

D.2 ROBUSTNESS TO LINEAR PROBABILITY MODEL FOR INDICATOR OUTCOMES

D.3 ROBUSTNESS TO OUTLIERS IN THE DATA

D.4 ROBUSTNESS TO MATCHING ALGORITHM

E EFFECT HETEROGENEITY ALONG THE AGE DISTRIBUTION

F EFFECT HETEROGENEITY ALONG THE INCOME DISTRIBUTION